OD

OPTIDENT DENTAL CLINIC MANAGEMENT SYSTEM

Full-stack Healthcare Platform (Node.js • Express • MongoDB Native)

MASTER EXPLORATORY TESTING CHARTER

Comprehensive Sprint Backlog Verification & Session-Based Suite

This master exploratory testing charter serves as the primary system-level quality assurance roadmap for the OptiDent Dental Clinic Management System. Structured in rigorous alignment with the Agile Sprint Backlog (comprising 57 User Stories across Super Admin, Doctor, Receptionist, and Client/Patient roles), this document establishes structured exploration missions, heuristic attack vectors, data prerequisites, boundary conditions, and formal bug classification protocols under Session-Based Exploratory Testing (SBET) methodology.

DOCUMENT SPECIFICATIONS & SPRINT METADATA

Target Application:	OptiDent Dental Management System (v1.0 Production Release)
Testing Methodology:	Session-Based Exploratory Testing (SBET) & Heuristic Risk Analysis
Sprint Backlog Scope:	All 57 User Stories: Super Admin (18), Doctor (13), Receptionist (15), Client (11)
Target Audience:	Software Quality Assurance Engineers, Systems Analysts, Developers, Stakeholders
Test Execution Period:	November 2026 Sprint Quality Gate
Environment & Ports:	Node.js v24.12.0 • Native MongoDB 127.0.0.1:27017/OptiDent • Local:3000
Third-Party Integrations:	Google Calendar API Service Account • Nodemailer SMTP • Session Security

DOCUMENT REVISION & QUALITY SIGN-OFF

Rev	Date	Prepared By	Role	Status
v1.0	Nov 09, 2026	QA Engineering Lead	Systems Quality Lead	**APPROVED FOR EXECUTION**
v1.1	Nov 09, 2026	Lead Systems Analyst	Systems Analyst / Architect	**VERIFIED AGAINST SPRINT**

SECTION 1: TESTING METHODOLOGY & SBET FRAMEWORK

Session-Based Exploratory Testing (SBET) is an evidence-backed testing method that combines spontaneous human curiosity, critical reasoning, and heuristic attack patterns with structured time-boxing and formal charter metrics. Rather than following rigid pre-scripted steps that miss subtle state corruptions and edge cases, testers operate under targeted "Missions" designed to expose high-concurrency collisions, boundary flaws, role-based leaks, and clinical safety issues.

1.1 Session Time-Box & TBS Metric Allocation

Each exploratory session is strictly time-boxed to prevent tester fatigue and maintain intense cognitive focus. Uninterrupted session blocks are monitored using the TBS Breakdown formula:

Session Class	Time-box	Test Execution (T)	Bug Investigation (B)	Setup & Logging (S)
Short Session	30 Minutes	20 min (65%)	6 min (20%)	4 min (15%)
Standard Session	60 Minutes	40 min (65%)	12 min (20%)	8 min (15%)
Extended Session	90 Minutes	60 min (65%)	18 min (20%)	12 min (15%)

1.2 Applied Quality Heuristics (Attack Toolset)

CORE HEURISTIC FRAMEWORKS UTILIZED IN TEST MISSIONS

- SFDIPOT (San Francisco Depot): Structure (Express/Node/Mongo stack), Function (booking, charting, vault), Data (BSON schema, timestamps), Interfaces (admin, doctor, receptionist, client), Platform (Windows, modern browsers), Operations (reception rush, simultaneous booking), Time (Manila PHT timestamps, slot collision).
- Goldilocks Heuristic: Input testing with values that are Too Big (overflow notes, long names), Too Small (empty strings, zero amounts, single-char passwords), and Just Right (valid dental medical payloads).
- CRUD Matrix & Inverted RBAC: Exercising Create, Read, Update, Delete lifecycles across every entity while deliberately swapping bearer sessions to trigger unauthorized privilege escalation.
- Concurrency & Race Condition Probing: Simulating two independent actors claiming the exact same operatory chair or doctor time slot at the exact millisecond.

1.3 Defect Severity & Clinical Risk Classification

Level	Classification	Clinical & Technical Impact	SLA Response
P1	**CRITICAL BLOCKER**	Patient safety hazard (e.g., missing medical allergy alert), data loss, appointment collision, or security breach.	Immediate (< 2 hrs)
P2	**MAJOR DEFECT**	Core user story broken without workaround (e.g., unable to approve appointments, vault restore failing, Google sync broken).	High (< 8 hrs)
P3	**MODERATE DEFECT**	Feature functional but with friction, non-blocking UI glitch, formatting errors in audit logs or reports.	Medium (< 24 hrs)
P4	**LOW / COSMETIC**	Minor cosmetic misalignment, typos, minor layout spacing, non-essential label polish.	Scheduled Sprint

SECTION 2: TEST ENVIRONMENT, CREDENTIALS & SETUP REQUIREMENTS

To maintain reproducible exploratory testing conditions, all sessions must be executed against the designated test environment pre-seeded with multi-role accounts, clinic operating hours, dental services, and appointment lifecycles.

2.1 Master Test Account Credentials Matrix

Role	Username / Email	Default Password	Pre-configured State & Purpose
Super Admin	optidentt@gmail.com	Opident1014	Master admin rights, access to vault, clinic controls, audit trail.
Doctor (Primary)	doctor@optident.com	DoctorPass#2026	Assigned practitioner, clinical charting, odontogram, approvals.
Doctor (Secondary)	dr.santos@optident.com	SantosSecure#1	Secondary provider for schedule collision & transfer testing.
Receptionist	reception@optident.com	FrontDesk#2026	Front-desk triage, quick walk-in registration, live queue management.
Client (Existing)	oniichanejmn@gmail.com	ClientSecret#1	Active patient with existing clinical history and booked visits.
Client (New)	walkin.patient@test.com	PatientNew#2026	Fresh account for self-service onboarding and first-time booking.
Soft-Deleted User	archived.doc@optident.com	[Deactivated]	Pre-soft-deleted user stored in Recovery Vault for restoration tests.
Locked User	locked.user@optident.com	Invalid3xAttempts	Account locked via SA-010 security threshold for unlock testing.

2.2 System Configuration & Environmental Prerequisites

ENVIRONMENTAL PARAMETERS & INTEGRATION SETUP

- Clinic Baseline Operating Hours: Monday to Saturday, 08:00 AM – 05:00 PM; Lunch Break Blackout: 12:00 PM – 01:00 PM; Sunday: Closed.
- Operatory Facilities: Operatory Chair 1 (General Dentistry), Operatory Chair 2 (Pediatric/Prostho), Operatory Chair 3 (Oral Surgery).
- Google Calendar API: Service Account active (optident-calendar@master-host-514202-g2.iam.gserviceaccount.com) with event insert/delete scopes.
- Local Server & Database: Node.js v24.12.0 running on localhost:3000 connected to MongoDB native at mongodb://127.0.0.1:27017/OptiDent.
- Browser Targets: Google Chrome v131+, Microsoft Edge v131+ (minimum 1280x800 resolution for desktop; 375x667 for mobile responsive tests).

SECTION 3: SPRINT TEST CHARTERS: SUPER ADMIN ROLE

MISSION 1: IDENTITY GOVERNANCE, RBAC & INVITATION PIPELINE

ROLE: SUPER ADMIN | TIMEBOX: 60 Minutes | STORIES: SA-001, SA-002, SA-003, SA-004, SA-005

Charter Objective & Scope: Verify administrative authentication, tokenized email invitations for doctors/receptionists, multi-user role assignment, and unbreakable role boundary enforcement across all endpoints.

EXPLORATION SCOPE & HIGH-RISK AREAS

- Multi-user CRUD operations across super_admin, doctor, receptionist, and patient roles.
- Invitation link generation: Token randomness, expiry timestamps, and one-time use consumption.
- Enforcement of Master Administrator rights: Protection against self-demotion or accidental deletion of the primary super admin.

HEURISTIC ATTACK VECTORS & STRESS SCENARIOS

- Brute force / credential stuffing against SA-001 login endpoint to verify lockout trigger.
- Manipulate invitation URL parameters to force role elevation (e.g., register as receptionist but spoof role=super_admin).
- Direct API tampering: Send HTTP requests to /api/admin/ endpoints using a client/patient session cookie (IDOR attack).

EXPECTED SYSTEM BEHAVIOR & FAILURE SYMPTOMS

- Expected: Super Admin retains full governance; unauthorized roles receive HTTP 401/403 with zero data disclosure.
- Failure Symptoms: Role leakage, privilege escalation, reusable invitation tokens, or unhandled 500 errors.

MISSION 2: DATA LIFECYCLE, SOFT-DELETE RECOVERY VAULT & ARCHIVAL

ROLE: SUPER ADMIN | TIMEBOX: 60 Minutes | STORIES: SA-006, SA-007, SA-013

Charter Objective & Scope: Evaluate soft-deletion resilience across users, appointments, and invitations. Verify that records can be inspected in the Data Recovery Vault, restored cleanly without orphaned references, or permanently purged.

EXPLORATION SCOPE & HIGH-RISK AREAS

- Soft-delete execution on patient records, doctor accounts, and booked appointments.
- Data Recovery Vault UI (SA-006): Tabular rendering, collection filtering (All, Users, Invitations, Appointments).
- Restore mechanics: Restoring a user must instantly reactivate login capability and restore linked relationships.
- Database Backup & Export (SA-013): JSON/BSON database dumps, backup file validity, and archival retention.

HEURISTIC ATTACK VECTORS & STRESS SCENARIOS

- Attempt to log in with a soft-deleted user account (must be blocked with deactivated account warning).
- Search for soft-deleted patients in the Receptionist search bar (must NOT appear in active search results).
- Purge a record permanently, then attempt to restore via forged API calls to test backend purge finality.

EXPECTED SYSTEM BEHAVIOR & FAILURE SYMPTOMS

- Expected: Soft-deleted items vanish from active interfaces, appear in Recovery Vault, and restore/purge flawlessly.
- Failure Symptoms: Orphaned appointments, "Error connecting to Recovery Vault", hard delete without vault entry.

MISSION 3: CLINIC CONFIGURATION, OPERATING HOURS & MAINTENANCE LOCK

ROLE: SUPER ADMIN | TIMEBOX: 45 Minutes | STORIES: SA-008, SA-009, SA-010, SA-011, SA-012

Charter Objective & Scope: Validate clinic profile updates, operating hours customization, lunch break blackout enforcement, session security limits, and system-wide maintenance mode restrictions.

EXPLORATION SCOPE & HIGH-RISK AREAS

- Clinic Information Settings (SA-008): Update clinic name, telephone, emergency mobile, and Fortich St. address.
- Operating Hours (SA-009): Adjust daily opening/closing hours and configure 12:00 PM - 01:00 PM lunch blackout.
- Security Settings (SA-010): Max failed attempts, idle session timeout enforcement.
- Maintenance Mode (SA-011): Enable system-wide maintenance lock; verify client/doctor access cutoff.
- Integration Controls (SA-012): Google Service Account toggle, SMTP email credentials validation.

HEURISTIC ATTACK VECTORS & STRESS SCENARIOS

- Alter operating hours to end at 03:00 PM; verify patient booking modal instantly disables 03:00 PM - 05:00 PM slots.
- Book an appointment during the 12:00 PM - 01:00 PM lunch break via direct API call to test backend schedule enforcement.
- Activate Maintenance Mode while an active patient is filling a booking form; verify graceful redirection to maintenance notice.

EXPECTED SYSTEM BEHAVIOR & FAILURE SYMPTOMS

- Expected: Operating hours changes instantly dictate slot availability; maintenance mode locks non-admin sessions safely.
- Failure Symptoms: Booking allowed outside operating hours, super admin locked out during maintenance, unhandled crashes.

MISSION 4: FORENSIC AUDIT TRAIL & BUSINESS INTELLIGENCE ANALYTICS

ROLE: SUPER ADMIN | TIMEBOX: 60 Minutes | STORIES: SA-014, SA-015, SA-016, SA-017, SA-018

Charter Objective & Scope: Verify real-time audit logging accuracy in Philippine Standard Time (Asia/Manila), actor attribution, removal of redundant columns, and generation of accurate patient, appointment, cancellation, and sales financial reports.

EXPLORATION SCOPE & HIGH-RISK AREAS

- Administrative Audit Trail (SA-017): Manila timestamp formatting, actor email, colored role badge, IP address, and human-readable Event Details (e.g., LOGIN_SUCCESS, CREATE_USER, RESTORE_RECORD).
- Appointment Reports (SA-015) & Cancellation Reports (SA-016): Verified against MongoDB database totals.
- Sales & Financial Reports (SA-018): Ledger reconciliation of consultation fees and completed dental procedure totals.
- Data Export Integrity: Verify CSV and PDF export outputs against live dashboard tables.

HEURISTIC ATTACK VECTORS & STRESS SCENARIOS

- Perform 5 rapid administrative actions (create user, soft delete, restore, alter hours, login) and audit the log ordering.
- Search audit trail by actor email, action type, and IP address to verify real-time client-side filter integrity.
- Verify that no event logs display cryptic raw JSON or "[object Object]" in the Event Details column.

EXPECTED SYSTEM BEHAVIOR & FAILURE SYMPTOMS

- Expected: Audit logs are timestamped accurately in Asia/Manila with full actor attribution; financial reports balance to the cent.
- Failure Symptoms: Timestamps in UTC, missing actor emails, undefined fields, mathematical discrepancies in report sums.

SECTION 4: SPRINT TEST CHARTERS: DOCTOR ROLE

MISSION 5: CLINICAL HISTORY, INTERACTIVE ODONTOGRAM & TREATMENT CHARTING

ROLE: DOCTOR | TIMEBOX: 75 Minutes | STORIES: D-001, D-002, D-003, D-004, D-005, D-006

Charter Objective & Scope: Explore doctor onboarding via invitation token, clinical history inspection, surface-by-surface odontogram charting, procedure logging, materials documentation, and prescription creation.

EXPLORATION SCOPE & HIGH-RISK AREAS

- Account activation via email invitation token (D-001) and secure doctor authentication (D-002).
- Patient Medical History Access (D-003): Instant visibility of high-risk medical conditions and drug allergies.
- Interactive Odontogram (D-005): 32-tooth FDI/Universal notation, surface selection (Mesial, Distal, Occlusal, Incisal, Facial, Lingual).
- Procedure Recording (D-006): Treatment categorization, materials used, procedure cost calculations, and progress notes.

HEURISTIC ATTACK VECTORS & STRESS SCENARIOS

- Select multiple tooth surfaces on the odontogram and save; reload page to verify surface-level persistence in MongoDB.
- Submit procedure notes containing special characters, medical symbols, and formatted line breaks to test sanitization.
- Attempt to edit a completed clinical record from a past date to test historical record immutability rules.

EXPECTED SYSTEM BEHAVIOR & FAILURE SYMPTOMS

- Expected: Odontogram renders accurately; medical allergies trigger visible clinical warnings; records save without data corruption.
- Failure Symptoms: Odontogram surfaces failing to persist, XSS execution in progress notes, missing allergy alerts.

MISSION 6: SCHEDULE GOVERNANCE, CONSULTATION PIPELINE & ACTIONS DROPDOWN

ROLE: DOCTOR | TIMEBOX: 60 Minutes | STORIES: D-007, D-008, D-009, D-010, D-016, D-017, D-018

Charter Objective & Scope: Validate doctor working hours configuration, appointment approval/decline workflows with mandatory explanation logging, consultation lifecycle progression, and the unified Actions dropdown menu.

EXPLORATION SCOPE & HIGH-RISK AREAS

- Appointment Review (D-016, D-017): Approve pending appointment and verify instant status change to "Scheduled".
- Appointment Decline (D-018): Decline booking request and verify that mandatory explanation is logged and emailed to patient.
- Actions Dropdown: Verify responsive floating viewport positioning for [Start Consultation], [Add to Google Calendar], [Complete & Chart Direct], [Dental History], [Cancel Appointment].
- Lifecycle Status Progression: Pending -> Scheduled -> In Process -> Completed.

HEURISTIC ATTACK VECTORS & STRESS SCENARIOS

- Decline an appointment with an empty explanation string to ensure form validation prevents submission.
- Click Actions dropdown near the bottom edge of the browser viewport to test dropdown collision/clipping behavior.
- Transition an appointment directly from "Pending" to "Completed" to verify state machine integrity.

EXPECTED SYSTEM BEHAVIOR & FAILURE SYMPTOMS

- Expected: Actions dropdown operates smoothly without UI clipping; approvals and declines update calendars immediately.
- Failure Symptoms: Side-by-side button spill, untracked cancellations, missing decline rationale.

SECTION 5: SPRINT TEST CHARTERS: RECEPTIONIST ROLE

MISSION 7: WALK-IN QUICK REGISTRATION, REAL-TIME SEARCH & QUEUE TRIAGE

ROLE: RECEPTIONIST | TIMEBOX: 60 Minutes | STORIES: R-001, R-002, R-003, R-004, R-005, R-006, R-009

Charter Objective & Scope: Evaluate front-desk walk-in patient quick onboarding, high-speed substring patient search, live waiting room queue triage, doctor chair assignment, and emergency medical clearance checks.

EXPLORATION SCOPE & HIGH-RISK AREAS

- Quick Walk-in Registration (R-003): Fast registration without requiring patient email verification or portal password.
- Real-time Patient Search (R-004): Instant search by first name, last name, phone number, and patient ID.
- Queue Management (R-006): Waiting room queue ordering, status transitions (Waiting in Lobby -> In Chair -> Completed).
- Patient Medical Access (R-009): Reviewing medical clearance alerts prior to seating patient in operatory.

HEURISTIC ATTACK VECTORS & STRESS SCENARIOS

- Quick-register a patient with an existing phone number to test duplicate detection and profile merging logic.
- Perform search queries with 1 character, special regex characters (.*, \d), and non-Latin names to test search resilience.
- Reorder queue priorities dynamically and verify real-time synchronization on the doctor dashboard queue view.

EXPECTED SYSTEM BEHAVIOR & FAILURE SYMPTOMS

- Expected: Fast walk-ins complete in under 30 seconds; search results return in < 100ms; queue stays in strict synchronization.
- Failure Symptoms: Duplicate patient records, search input freezing browser, queue state discrepancies between desks.

MISSION 8: APPOINTMENT DISPATCH, RESCHEDULING & CONFLICT ELIMINATION

ROLE: RECEPTIONIST | TIMEBOX: 60 Minutes | STORIES: R-008, R-010, R-011, R-012, R-013, R-014, R-015, R-019

Charter Objective & Scope: Test front-desk appointment creation, schedule conflict prevention across multiple dentist calendars, appointment rescheduling, and receptionist self-service password recovery via 6-digit OTP.

EXPLORATION SCOPE & HIGH-RISK AREAS

- Appointment Schedule Monitoring (R-008): Multi-doctor calendar grid, chair occupancy visualization.
- Appointment Creation (R-013): Phone-in bookings with instant slot locking.
- Rescheduling Dynamics (R-011, R-014): Moving dates and times while verifying automated conflict detection.
- Cancellations (R-010): Logging cancellation reasons and releasing chair slots for immediate rebooking.
- OTP Password Reset (R-019): Requesting 6-digit OTP via email and resetting receptionist credentials.

HEURISTIC ATTACK VECTORS & STRESS SCENARIOS

- Attempt to reschedule an existing appointment to a slot where the doctor already has an active consultation.
- Book two walk-in appointments for the exact same operatory chair at the exact same time.
- Submit an expired or tampered OTP token to verify brute-force defense on the password reset route.

EXPECTED SYSTEM BEHAVIOR & FAILURE SYMPTOMS

- Expected: Conflicting bookings are cleanly rejected with clear user messaging; vacated slots become instantly bookable.
- Failure Symptoms: Overlapping appointments on the same doctor/chair, silent booking failures, OTP token replay.

SECTION 6: SPRINT TEST CHARTERS: CLIENT / PATIENT ROLE

MISSION 9: SELF-SERVICE ONBOARDING, MEDICAL DISCLOSURE & ACCOUNT SECURITY

ROLE: CLIENT / PATIENT | TIMEBOX: 60 Minutes | STORIES: C-001, C-002, C-003, C-007, C-008, C-011

Charter Objective & Scope: Verify patient portal registration, comprehensive health questionnaire intake, personal profile customization, medical history access, and OTP-driven password recovery.

EXPLORATION SCOPE & HIGH-RISK AREAS

- Self-Registration (C-001) & Login (C-002): Strong password policy enforcement, email verification.
- Patient Onboarding (C-003): Health questionnaire covering cardiac issues, diabetes, bleeding disorders, and drug allergies.
- Medical History Portal (C-007): Patient-facing clinical summaries, past procedures, prescription instructions, and receipts.
- Profile Customization (C-008): Updating phone number, home address, emergency contacts, profile photo upload.
- Verification & Password Reset (C-011): 6-digit email OTP delivery, countdown timer, and secure password update.

HEURISTIC ATTACK VECTORS & STRESS SCENARIOS

- Upload oversized image files (> 10MB) or invalid file extensions (.exe, .svg) to profile picture upload.
- Submit health questionnaire with all risk factors checked; verify that high-risk medical alerts display on doctor chart.
- Request 5 consecutive OTP reset tokens in 60 seconds to test rate-limiting safeguards.

EXPECTED SYSTEM BEHAVIOR & FAILURE SYMPTOMS

- Expected: Intake medical data transfers accurately to doctor charts; passwords and personal data remain protected.
- Failure Symptoms: Unsanitized file uploads, medical risks failing to notify doctor, lack of OTP rate limiting.

MISSION 10: ONLINE SELF-BOOKING, DYNAMIC SLOTS, GOOGLE CALENDAR & QR PASS

ROLE: CLIENT / PATIENT | TIMEBOX: 75 Minutes | STORIES: C-004, C-005, C-006, C-009, C-010

Charter Objective & Scope: Explore online dental appointment booking, real-time slot conflict prevention, appointment lifecycle tracking, automated email notifications, Google Calendar 1-Click sync, Service Account push sync, .ics file download, and digital QR Pass generation.

EXPLORATION SCOPE & HIGH-RISK AREAS

- Online Self-Booking Wizard (C-004): Service selection (Consultation, Scaling, Extraction, Surgery), attending doctor selection, interactive date/time picker.
- Dynamic Slot Availability (C-005): Real-time blackout of booked slots, past dates, and lunch break (12:00 PM - 01:00 PM).
- Appointment Lifecycle Tracking (C-006): Real-time status badge updates (Pending Approval -> Scheduled -> Completed / Cancelled).
- Notifications Engine (C-009): Automated confirmation and schedule change emails.
- Calendar, Time/Date & QR Pass Support (C-010):
- a) 1-Click Google Calendar button prefilling event title, clinic location, dentist, and appointment time.
- b) Automatic zero-click background sync to patient/doctor calendars via Service Account (optident-calendar@master-host-514202-g2.iam.gserviceaccount.com).
- c) Downloadable .ics file for Apple Calendar, Outlook, and offline calendar import.
- d) Dynamic QR Pass rendering containing encoded Appointment ID, Patient Name, and Time for contact-free reception check-in.

HEURISTIC ATTACK VECTORS & STRESS SCENARIOS

- Open two browser windows with two different patient accounts selecting the same doctor and slot; click "Confirm Booking" simultaneously to test concurrency race condition.
- Book an appointment, verify 1-Click Google Calendar link formatting, then cancel the appointment and verify removal across calendars.
- Download .ics file and verify iCalendar RFC 5545 compliance (UID, DTSTART in Asia/Manila, LOCATION, SUMMARY).

EXPECTED SYSTEM BEHAVIOR & FAILURE SYMPTOMS

• Expected: Simultaneous bookings cleanly resolve without double-booking; calendar invitations and QR passes generate instantly.

• Failure Symptoms: Double bookings, timezone skew in Google Calendar (e.g., UTC offset shift), corrupt .ics downloads.

SECTION 7: CROSS-CUTTING SECURITY, STRESS & HEURISTIC ATTACK VECTORS

Beyond individual functional roles, exploratory testing must stress the systemic seams, concurrency thresholds, and authorization boundaries that automated unit tests frequently overlook.

7.1 High-Risk Attack Vectors & Threat Matrix

Attack Vector	Target Vulnerability	Heuristic Test Technique	Severity
RBAC Elevation & IDOR	Direct API access to /api/admin/ endpoints by patient sessions.	Session swapping, header tampering with forged roles in cookies.	**P1 - Critical**
Double-Booking Race Condition	Simultaneous HTTP POST /api/client/book_appointment for same slot.	Asynchronous curl / fetch bursts firing within 5ms window.	**P1 - Critical**
Stored XSS in Clinical Notes	Cross-site scripting in chief complaints, notes, and odontogram.	Inject script tags (<script>alert(1)</script>) into clinical textareas.	**P2 - Major**
Soft-Delete Data Leakage	Deactivated users/records leaking into active search or reports.	Inspect JSON payloads of active queries for is_deleted: true flags.	**P2 - Major**
Audit Trail Tampering	Modification or deletion of historical audit log records.	Attempt direct POST/DELETE calls against audit log database collections.	**P1 - Critical**
Timezone Drift & Cal Skew	Appointments recorded in UTC displaying incorrect Manila local times.	Verify appointment times across DST, UTC vs PHT (+08:00) offsets.	**P2 - Major**

7.2 Data Boundary & Goldilocks Input Testing Checklist

BOUNDARY VALUE TESTING MATRIX ACROSS FORMS

- Patient Full Name: Test 1 character ("A"), 150 characters, numbers, and special symbols (e.g., "Ma. Concepción de la Cruz-Ñoñez Jr.").
- Contact Number: Test valid 11-digit mobile (09171234567), landlines (02-8123-4567), incomplete numbers (0912), and letters ("CALLME").
- Consultation / Procedure Fees: Test zero values (0.00), negative amounts (-150.00), extreme values (999,999,999.00), and alphabetic strings.
- Odontogram Notes: Test maximum payload size (> 5,000 words), emojis, Unicode dental charting characters, and HTML markup.
- Date Boundaries: Attempt to book appointments on past dates (yesterday), 5 years into the future, and on recognized clinic holidays.

SECTION 8: SESSION EXECUTION PROTOCOL & DEFECT REPORTING STANDARDS

Every exploratory testing session must be formally documented using the Session Log Template. When anomalies, regressions, or unexpected behaviors are discovered, testers must file standardized Defect Reports adhering to the format below.

8.1 Session-Based Exploratory Testing (SBET) Session Log Template

Session Log Field	Field Description & Required Input Format
Session Identifier	Unique ID: e.g., SBET-OPT-20261109-DOC-M05-01
Charter Mission #	Designated Mission (e.g., Mission 5: Clinical History & Odontogram)
Assigned Tester(s)	Full name and QA role of the executing tester
Execution Date & Time	Timestamp of session start and conclusion in Philippine Time (PHT)
Time-box Duration	Total elapsed session time (e.g., 60 minutes)
TBS Metric Breakdown	Percentage distribution: e.g., 65% Test (T) / 20% Bug (B) / 15% Setup (S)
Environment Details	Browser, OS build, Node server commit SHA, MongoDB dataset state
Exploration Notes	Chronological log of areas explored, heuristics applied, and observations
Defects Identified	List of bug IDs logged during session (e.g., BUG-OPT-042)
Follow-up Questions	Open questions or clarification items requiring Systems Analyst input

8.2 Standardized Defect Report Format (Bug Classification)

MANDATORY FIELDS FOR EVERY QUALITY DEFECT TICKET

- Defect Title: Concise, unambiguous summary following [Role / Module] [Component] [Observed Failure].
- User Story Reference: Exact sprint story ID (e.g., D-005, SA-006, C-010).
- Severity & Priority: P1 (Critical), P2 (Major), P3 (Moderate), P4 (Minor / Cosmetic).
- Pre-conditions & Test Account: Exact user credentials, pre-existing patient record ID, and browser environment.
- Step-by-Step Reproduction: Minimal deterministic steps required to reproduce the failure reliably.
- Expected Behavior: Specification-backed system behavior defined in the sprint acceptance criteria.
- Actual Observed Behavior: Exact failure symptom, error messages, screenshot/video references, and console logs.
- Backend Log Evidence: Relevant terminal output, Node.js stack traces, and MongoDB document state before/after.

SECTION 9: COMPLETE SPRINT TRACEABILITY MATRIX (ALL 57 STORIES)

The following traceability matrix establishes direct 100% test coverage mapping from every single Sprint User Story to its designated exploratory testing charter mission, test focus, and acceptance verification criteria.

9.1 Super Admin User Stories (SA-001 through SA-018)

Story ID	Function Name	Sprint Story Description	Mission	Key Verification Criteria
SA-001	User Login	Secure login using registered admin credentials.	**Mission 1**	Auth token, session cookie, failed attempt lock.
SA-002	Doctor & Receptionist Invitation	Invite staff securely via tokenized email.	**Mission 1**	Unique token generation, SMTP delivery, one-time use.
SA-003	Multi-User Management	Manage Super Admin, Doctor, Receptionist, Client.	**Mission 1**	Full CRUD across user roles, active status toggling.
SA-004	Role-Based Access Control (RBAC)	Enforce role permissions strictly across endpoints.	**Mission 1**	Endpoint protection, forbidden redirection, zero leak.
SA-005	Master Administrator Rights	Full control over system settings, users, and data.	**Mission 1**	Override capabilities, immunity to self-deletion.
SA-006	Soft Delete	Deactivate accounts/records into Recovery Vault.	**Mission 2**	Recovery Vault UI, restore record, permanent purge.
SA-007	Data Archiving	Archive inactive users preserving historical data.	**Mission 2**	Historical data preservation, fast active queries.
SA-008	Clinic Information Settings	Update clinic details, contact info, and branding.	**Mission 3**	Real-time update across patient and staff dashboards.
SA-009	Operating Hours Settings	Configure working days, hours, and lunch breaks.	**Mission 3**	Booking calendar slot availability synchronization.
SA-010	System Security Settings	Configure session timeouts and lockout thresholds.	**Mission 3**	Session invalidation after timeout, brute-force defense.
SA-011	Maintenance Mode	Enable system maintenance lock for non-admin users.	**Mission 3**	Global lock screen for patients/doctors, admin bypass.
SA-012	System Integration Controls	Manage Google Service Account and SMTP controls.	**Mission 3**	API key validation, sync toggle, health check status.
SA-013	Database Backup and Export	Back up and export database collections securely.	**Mission 2**	JSON/BSON dump generation, archive download integrity.
SA-014	Patient Reports	Generate comprehensive patient demographic reports.	**Mission 4**	Registration growth, patient profile completeness.
SA-015	Appointment Reports	Generate reports for scheduled and completed visits.	**Mission 4**	Status breakdown, date filtering, doctor utilization.
SA-016	Cancellation Reports	Track canceled and declined visits with reasons.	**Mission 4**	Cancellation rationale logs, doctor decline metrics.
SA-017	System Activity Reports (Audit Trail)	Monitor administrative actions with PHT timestamps.	**Mission 4**	Asia/Manila time, actor email/role, clear event details.
SA-018	Sales & Financial Reports	Generate clinic sales, fees, and revenue reports.	**Mission 4**	Accurate revenue summation, billing itemization.

9.2 Doctor User Stories (D-001 through D-018)

Story ID	Function Name	Sprint Story Description	Mission	Key Verification Criteria
D-001	User Registration	Activate doctor account via invitation token link.	**Mission 5**	Token consumption, password setup, role binding.
D-002	User Login	Secure login to access clinical dashboard.	**Mission 5**	Doctor session validation, dashboard access.
D-003	Patient Medical History	View assigned patient medical background.	**Mission 5**	Medical alerts, drug allergies, health questionnaire.
D-004	Clinical Record Management	Record diagnoses, treatments, notes, procedures.	**Mission 5**	Clinical notes persistence, historical treatment log.
D-005	Diagnosis Recording	Interactive surface-by-surface tooth charting.	**Mission 5**	Odontogram 32-tooth rendering, condition saving.
D-006	Procedure Recording	Record dental procedures and materials used.	**Mission 5**	Procedure catalog, fee itemization, progress notes.
D-007	Doctor OTP Verification & Reset	Reset password securely via 6-digit email OTP.	**Mission 6**	OTP generation, email dispatch, expiry validation.
D-008	Doctor Profile Management	Update professional profile, license, consultation fee.	**Mission 6**	Profile fields persistence, public listing update.
D-009	Schedule Management	Manage working hours, shifts, and slot availability.	**Mission 6**	Doctor slot restriction, vacation blackout dates.
D-010	Appointment Status Tracking	Track status: Pending, Scheduled, In Progress, Done.	**Mission 6**	Pipeline state transitions, Actions dropdown trigger.
D-016	Appointment Approval	Approve incoming appointment booking requests.	**Mission 6**	Status changes to Scheduled, confirmation sent.
D-017	Appointment Confirmation & Sync	Confirm schedule and push to doctor Google Calendar.	**Mission 6**	Calendar event insertion via service account API.
D-018	Appointment Decline	Decline appointment requests with mandatory reason.	**Mission 6**	Mandatory reason input, patient decline notification.

9.3 Receptionist User Stories (R-001 through R-019)

Story ID	Function Name	Sprint Story Description	Mission	Key Verification Criteria
R-001	User Registration	Activate receptionist account via invitation email.	**Mission 7**	Token verification, receptionist role assignment.
R-002	User Login	Secure login to front-desk management portal.	**Mission 7**	Front-desk session creation, role permission check.
R-003	Patient Quick Registration	Quickly register walk-in patients efficiently.	**Mission 7**	Minimal fields, duplicate check, instant patient ID.
R-004	Patient Search	Search existing records by name, ID, or phone.	**Mission 7**	Sub-100ms substring search, zero UI freezing.
R-005	Patient Information Management	Update patient contact and registration details.	**Mission 7**	Emergency contact editing, audit trail logging.
R-006	Queue Management	Manage live waiting room patient order and status.	**Mission 7**	Waiting room queue ordering, attending doc assign.
R-008	Appointment Schedule Monitoring	View dentist schedules to prevent booking conflicts.	**Mission 8**	Multi-doctor schedule grid, chair occupancy check.
R-009	Patient Medical Info Access	Access necessary medical info for front-desk triage.	**Mission 7**	Pre-consultation clearance check, allergy flags.
R-010	Appointment Cancellation	Record cancellations and release appointment slots.	**Mission 8**	Slot freed on calendar, cancellation reason logged.
R-011	Appointment Rescheduling	Reschedule visits accommodating patient schedule.	**Mission 8**	Conflict prevention, doctor schedule re-allocation.
R-012	Receptionist Profile Management	Update receptionist account and profile info.	**Mission 8**	Profile update, contact details persistence.
R-013	Appointment Creation	Book phone-in and front-desk walk-in appointments.	**Mission 8**	Direct calendar booking, patient notification.
R-014	Appointment Rescheduling Controls	Manage schedule changes with conflict elimination.	**Mission 8**	Doctor availability verification, double-booking block.

Story ID	Function Name	Sprint Story Description	Mission	Key Verification Criteria
R-015	Appointment Management	View and modify all clinic appointment details.	**Mission 8**	Filter by status, date, doctor, or chair.
R-019	Receptionist OTP Password Reset	Reset password via 6-digit email OTP.	**Mission 8**	OTP verification, secure password replacement.

9.4 Client / Patient User Stories (C-001 through C-011)

Story ID	Function Name	Sprint Story Description	Mission	Key Verification Criteria
C-001	User Registration	Self-service account creation for patients.	**Mission 9**	Password strength validation, unique email rule.
C-002	User Login	Secure login to patient self-service dashboard.	**Mission 9**	Patient dashboard access, session authorization.
C-003	Self-Service Patient Onboarding	Enter personal profile and medical background.	**Mission 9**	Health intake form, allergy checklist, submit.
C-004	Online Self-Booking	Book dental appointment online based on schedules.	**Mission 10**	Interactive booking wizard, service/ doctor picker.
C-005	Real-Time Dynamic Slot Availability	Prevent slot clashing with live slot updates.	**Mission 10**	Real-time blackout of booked slots & lunch hour.
C-006	Appointment Lifecycle Tracking	Track status: Pending, Confirmed, Completed.	**Mission 10**	Visual status pills, transparent progress updates.
C-007	Comprehensive Medical History	Access authorized clinical records & prescriptions.	**Mission 9**	Read-only clinical summary, past receipts archive.
C-008	User Profile Customization	Update name, contact info, and profile avatar.	**Mission 9**	Profile photo upload, phone & address update.
C-009	Notifications Engine	Receive automated email updates on appointments.	**Mission 10**	Booking confirmation, doctor approval notice.
C-010	Calendar & QR Code Support	1-Click Google sync, auto sync, .ics & QR Pass.	**Mission 10**	Google Cal 1-Click, auto service push, .ics, QR.
C-011	Verification & Password Reset	Receive 6-digit email OTP for password recovery.	**Mission 9**	OTP delivery, 10-minute expiry, password update.

FORMAL QA SPRINT READINESS CERTIFICATION

This master exploratory testing charter covers 100% of all 57 sprint backlog user stories for the OptiDent Dental Clinic Management System. Execution of these 10 structured missions under Session-Based Exploratory Testing guarantees robust functional verification, deep security boundary probing, multi-calendar synchronization fidelity, and data recovery resilience prior to production deployment.